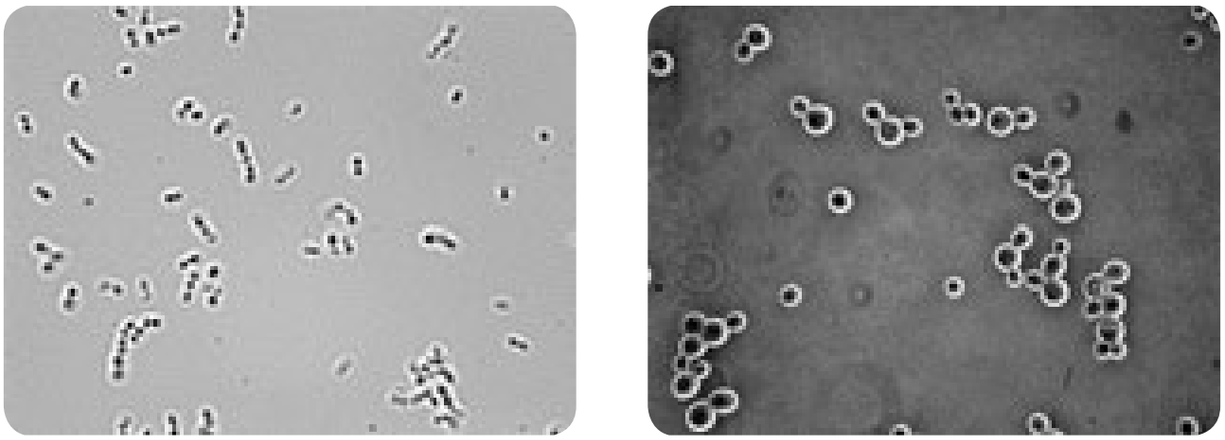


Figure 2.4: **(a)** *Saccharomyces* under bright field microscope **(b)** *Saccharomyces* under phase contrast microscope

- It is used to examine various intracellular components of living cells at relatively high resolution.
- It helps in studying cellular events such as cell division.
- It is used to visualize all types of cellular movements such as chromosomal and flagellar movements.

2.2 Fluorescence Microscope

Fluorescence microscope is a very powerful analytical tool that combines the magnifying properties of light microscope with visualization of fluorescence.

Fluorescence microscope is a type of light microscope which instead of utilizing visible light to illuminate specimens, uses a higher intensity (lower wavelength) light source that excites a fluorescence molecule called a fluorophore (also known as fluorochrome). Fluorescence is a phenomenon that takes place when the substances (fluorophore) absorbs light at a given wavelength and emits light at a higher wavelength. Thus, fluorescence microscopy combines the magnifying properties of the light microscope with fluorescence technology.

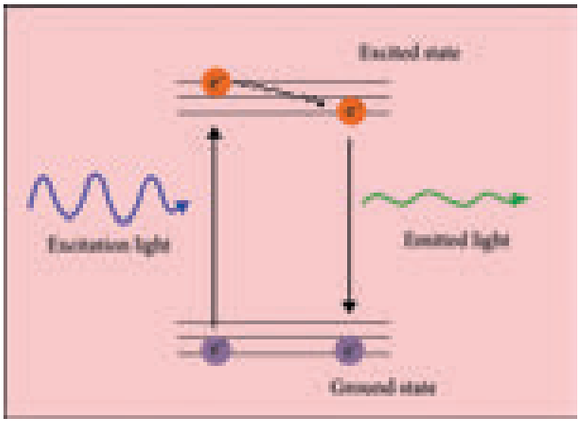


Figure 2.5: Principle of fluorescence

Infobits

British scientist Sir George G. Stokes first described fluorescence in 1852. He observed that the fluorophore emitted red light, when it was illuminated by ultraviolet excitation. Stokes noted that fluorescence emission always occurred at a longer wavelength than of the excitation light. This shift towards longer wavelength is known as stokes shift.

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